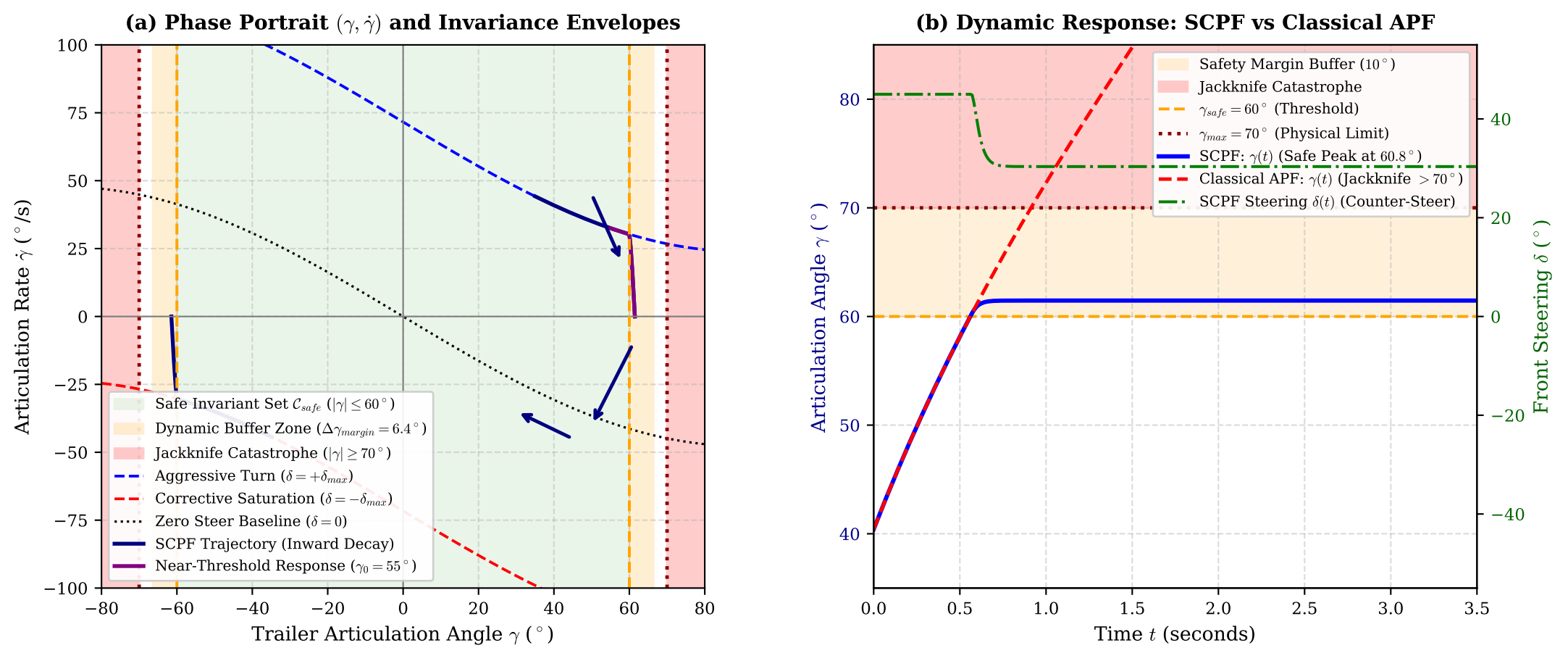

(a) Phase Portrait $(\gamma, \dot{\gamma})$ and Invariance Envelopes
Articulation Rate $\dot{\gamma}$ (°/s)
Trailer Articulation Angle γ (°)
Safe Invariant Set $\mathcal{C}_{safe}$ ($|\gamma| \leq 60°$)
Dynamic Buffer Zone ($\Delta\gamma_{margin} = 6.4°$)
Jackknife Catastrophe ($|\gamma| \geq 70°$)
Aggressive Turn ($\delta = +\delta_{max}$)
Corrective Saturation ($\delta = -\delta_{max}$)
Zero Steer Baseline ($\delta = 0$)
SCPF Trajectory (Inward Decay)
Near-Threshold Response ($\gamma_0 = 55°$)
(b) Dynamic Response: SCPF vs Classical APF
Articulation Angle γ (°)
Front Steering δ (°)
Time t (seconds)
Safety Margin Buffer (10°)
Jackknife Catastrophe
$\gamma_{safe} = 60°$ (Threshold)
$\gamma_{max} = 70°$ (Physical Limit)
SCPF: $\gamma(t)$ (Safe Peak at 60.8°)
Classical APF: $\gamma(t)$ (Jackknife $> 70°$)
SCPF Steering $\delta(t)$ (Counter-Steer)